

Olive Knot AI

Computer vision and plant research

National second place in Software
TÜBİTAK 2204-A, 2022

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Research question



Can a plant image help identify the visible stage of olive knot disease?

A CNN-based classifier grouped images into four stages. I put the predictions into a mobile research prototype.

Research prototype — not a certified diagnosis.

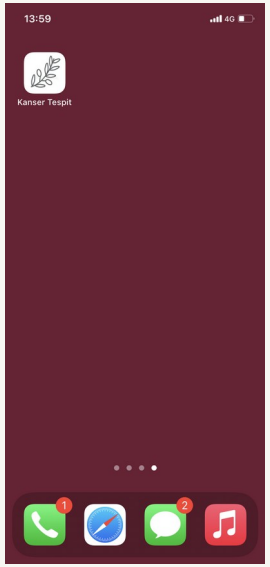
Mobile prototype



98.03% • stage 1

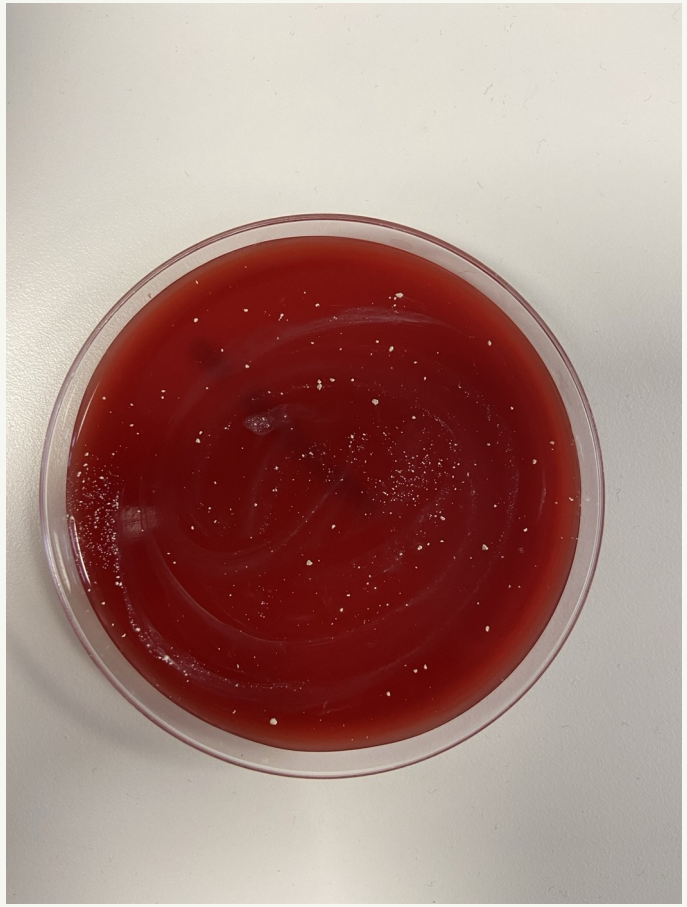


93.27% • stage 4



Confidence on these two images, not an overall accuracy measure.

Plant and laboratory study

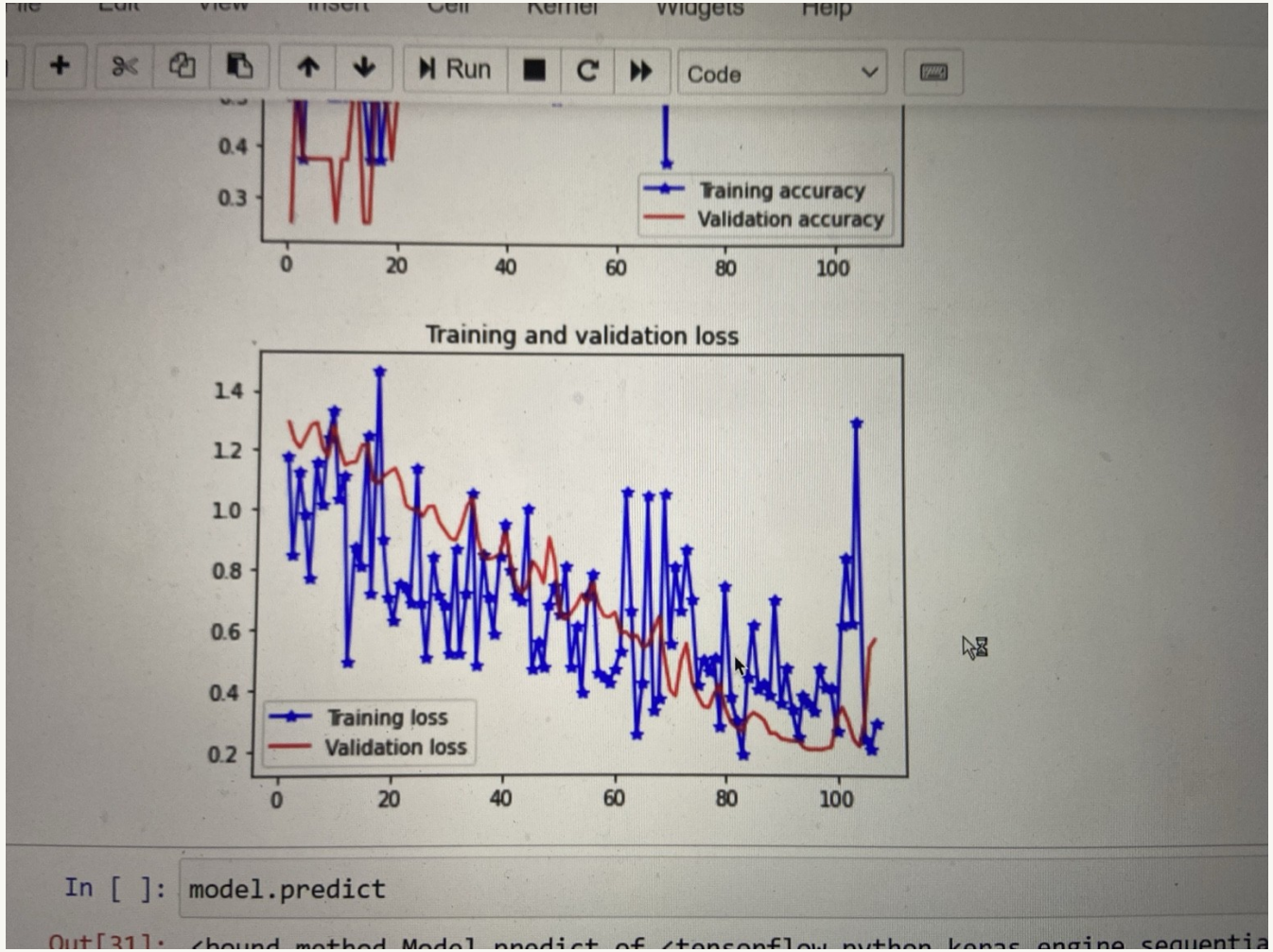


12 olive saplings

Four comparison groups
About four months of observation

The poster reports preliminary observations about disease progression and Bacillus subtilis. More samples and independent replication would be needed for a treatment claim.

Training and evaluation

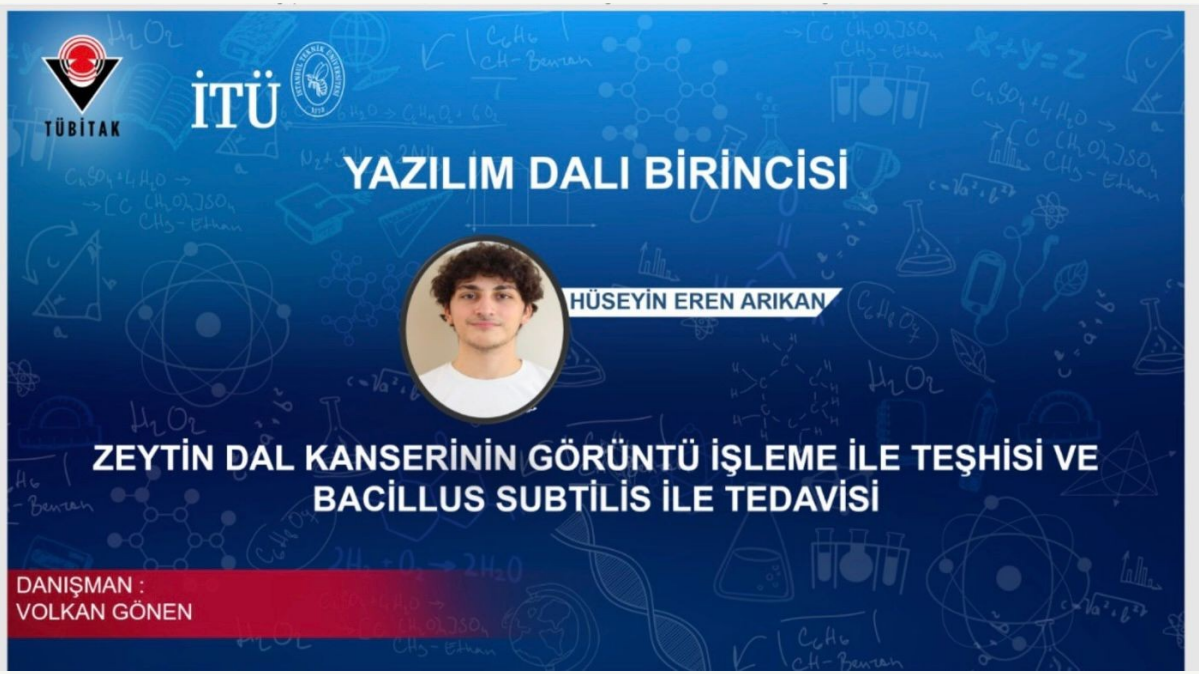


88.33%

classification accuracy reported in the 2022 poster

Self-reported research result. No independent field validation is claimed.

Regional and national recognition



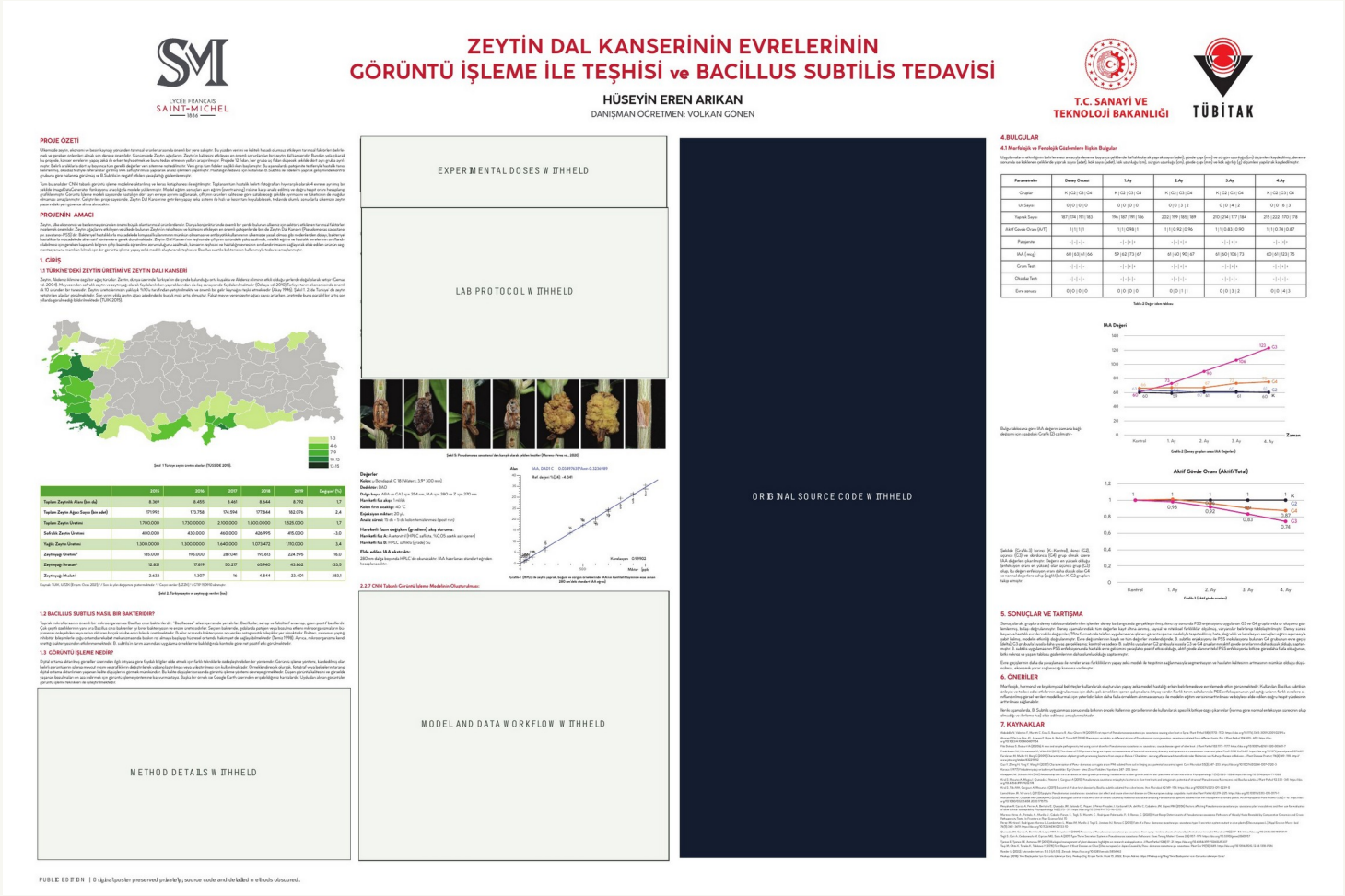
Regional first place

National second place in Software

TÜBİTAK 2204-A • 2022

25,095 project submissions and 37,283 students across all fields. National winners entered a further ISEF selection pathway.

Public edition of the original poster



The original layout and results remain. The public PDF removes code, exact doses and detailed methods.

Demonstrating the work



The project was built, presented and awarded.

The code, model weights, training data and original unredacted poster remain private.